

Data transforms

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Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase"))
```

General principles

- Make sure data is on the right scale for
- Visualization
- Statistical modeling

Load some data

We will use this expression set to look at how we use plots and tables to check for different characteristics

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bodymap_eset.RData")
load(file=con)
close(con)
bm = bodymap.eset
pdata=pData(bm)
edata=as.data.frame(exprs(bm))
fdata = fData(bm)
ls()
```

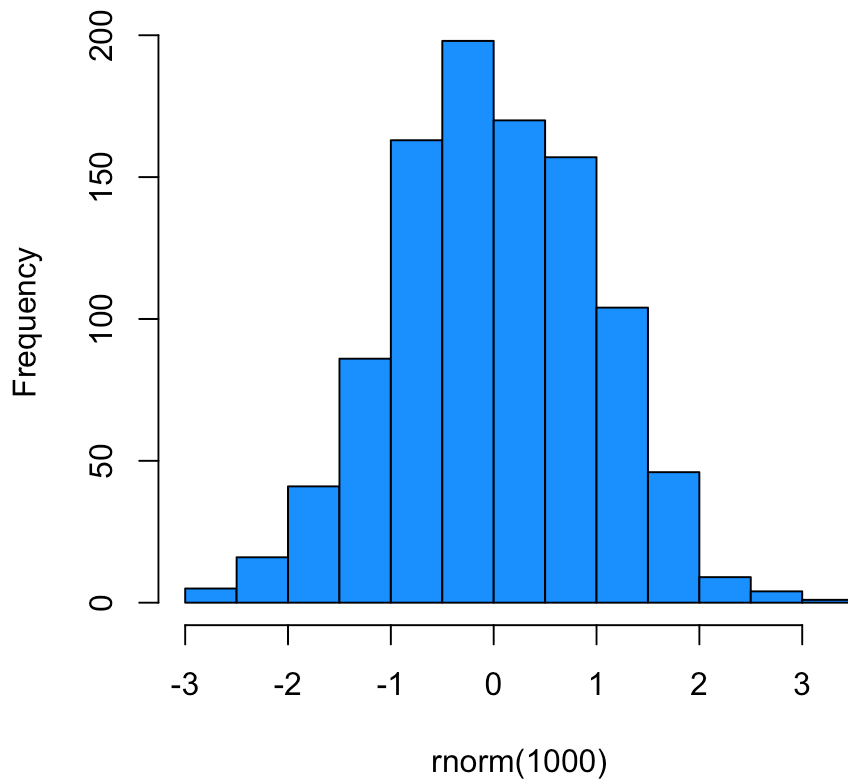
```
## [1] "bm"           "bodymap.eset" "con"           "edata"
## [5] "fdata"        "pdata"         "tropical"
```

Skewed distributions

We would like continuous data to be nice and symmetric like a normal distribution for two reasons: (1) plots are easier to see this way and (2) most statistical methods are designed to work better for non-skewed data.

```
hist(rnorm(1000),col=2)
```

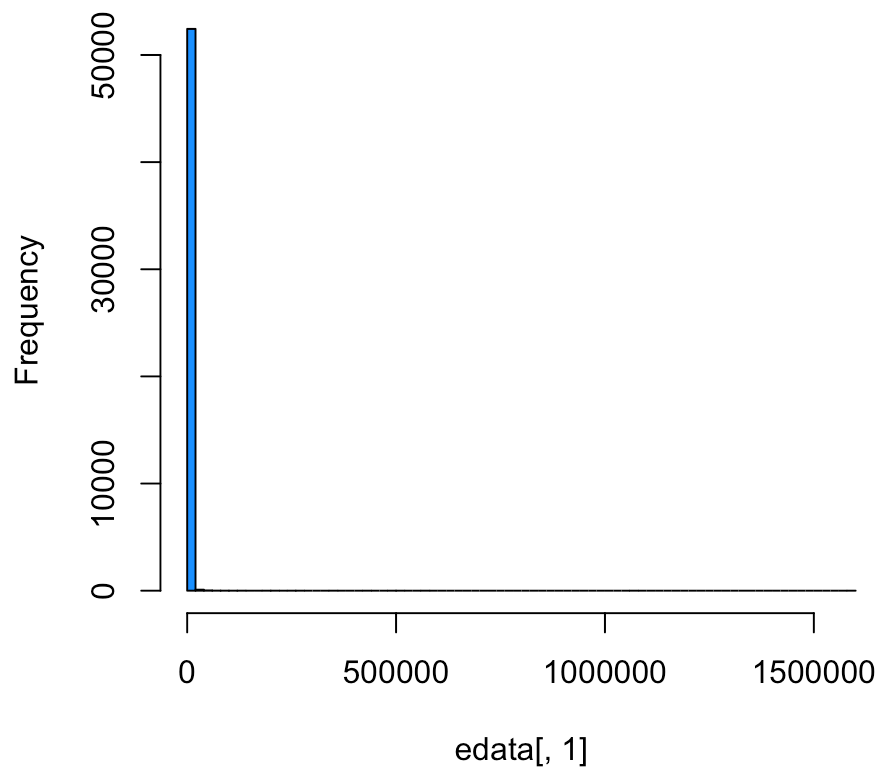
Histogram of rnorm(1000)



Realistically though most genomic data is skewed

```
hist(edata[,1],col=2,breaks=100)
```

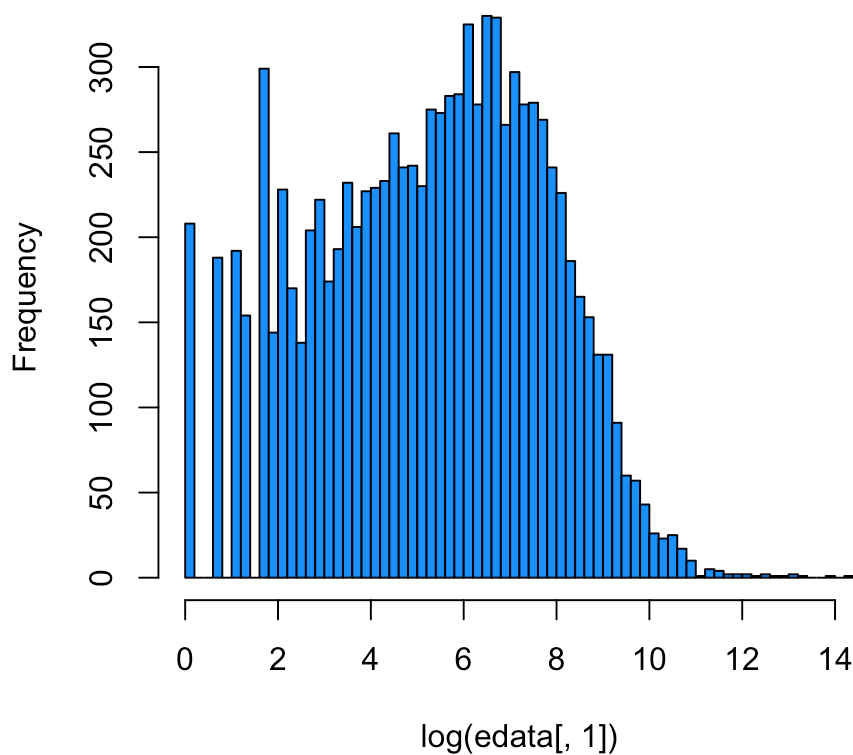
Histogram of edata[, 1]



One way to address this skew is to use a transformation. One common transformation is the `log` transform.

```
hist(log(edata[,1]),col=2,breaks=100)
```

Histogram of $\log(\text{edata[, 1]})$



One thing to be careful with is that values of zero become `-inf` when you apply the log transform because the log of zero isn't defined.

```
min(log(edata))
```

```
## [1] -Inf
```

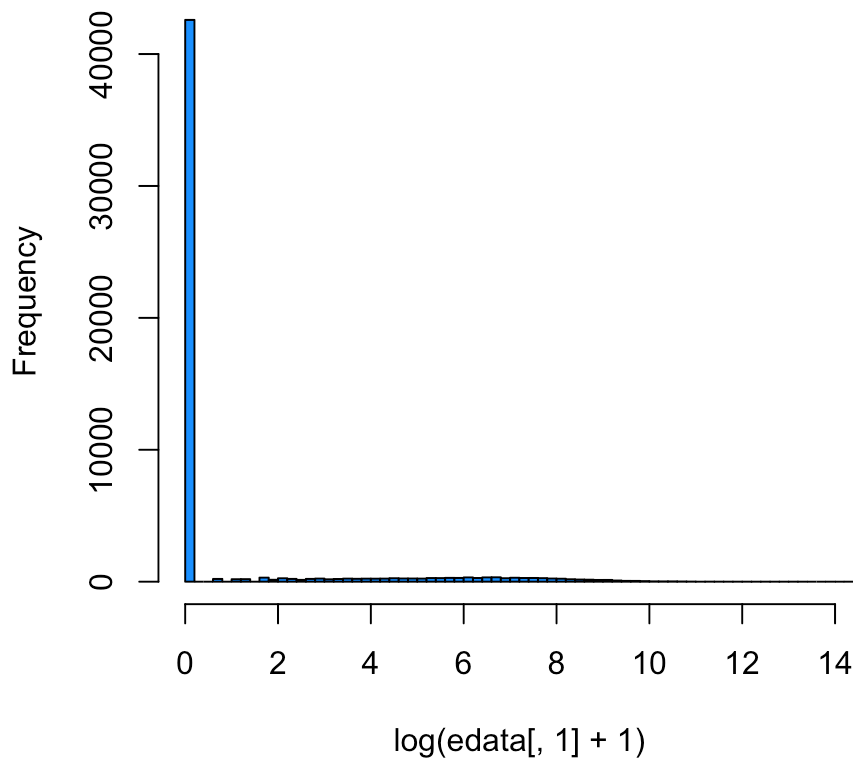
You can remove this problem by adding a small number to all the counts before taking the log. This doesn't change the overall distribution much but resolves the zero problem

```
min(log(edata[,1] + 1))
```

```
## [1] 0
```

```
hist(log(edata[,1] + 1),breaks=100,col=2)
```

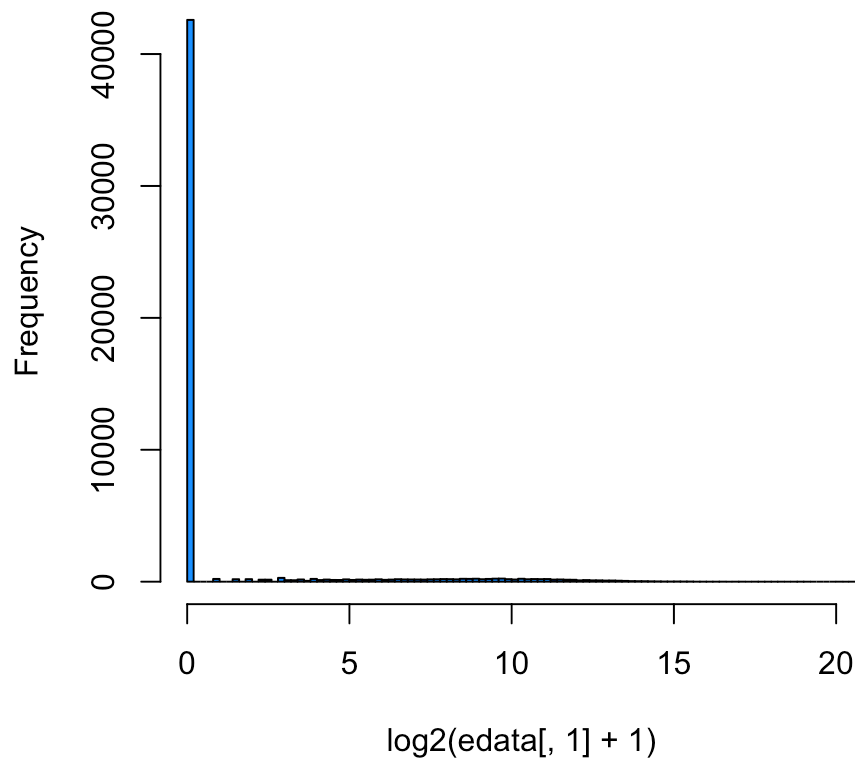
Histogram of $\log(\text{edata[, 1]} + 1)$



A common choice is to use the $\log_2$ transform. The reason is that data on the $\log_2$ scale can be expressed as fold changes (https://en.wikipedia.org/wiki/Fold_change). If $\log_2(x) - \log_2(y) = z$ then there is a z fold change difference between x and y .

```
hist(log2(edata[,1] + 1),breaks=100,col=2)
```

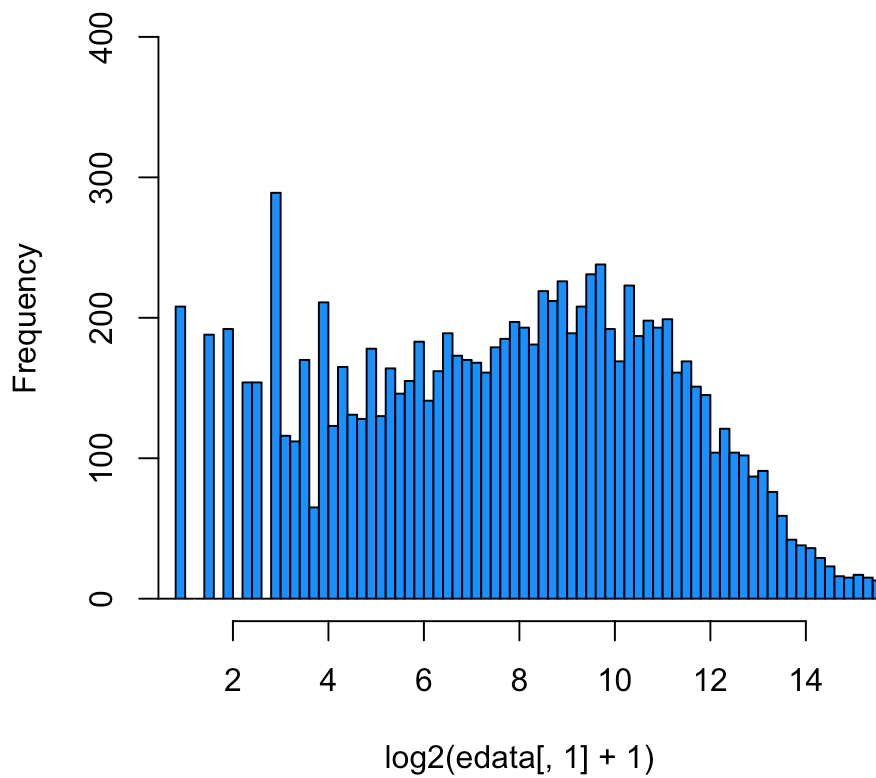
Histogram of $\log_2(\text{edata[, 1]} + 1)$



Let's zoom in on the values greater than zero (we'll come back to the zero values in a minute)

```
hist(log2(edata[,1] + 1),breaks=100,col=2,xlim=c(1,15),ylim=c(0,400))
```

Histogram of $\log_2(\text{edata[, 1]} + 1)$



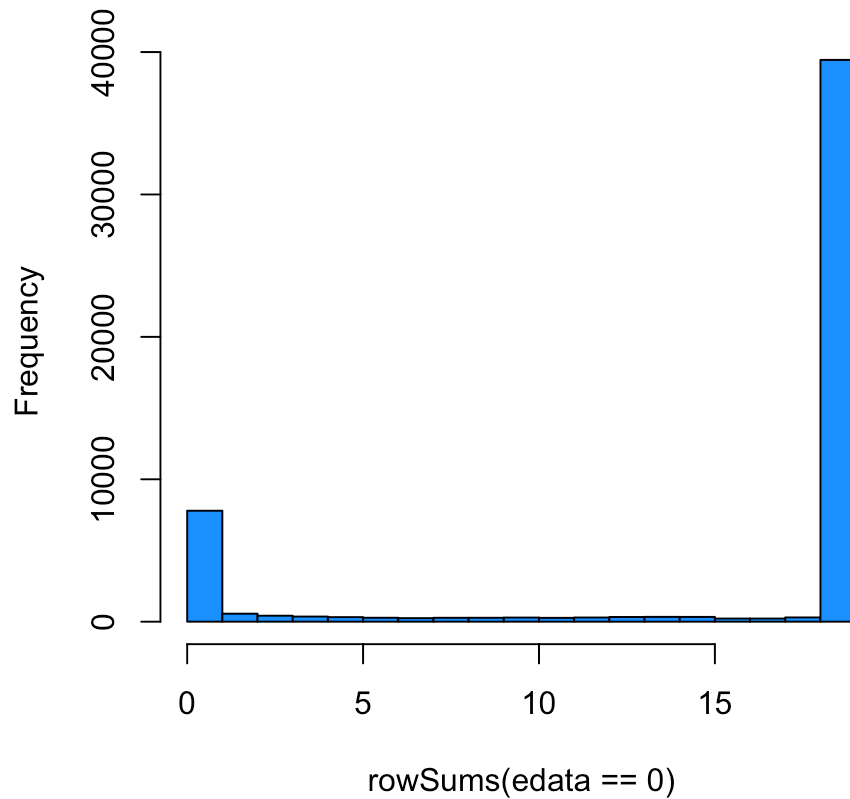
This is starting to look a little better but we still have the zeros problem.

Removing features without much data

In many types of sequencing data we observe that there is not much data for many features.

```
hist(rowSums(edata==0),col=2)
```

Histogram of rowSums(edata == 0)



A common pre-processing technique is to remove features that don't have much data. One way to identify and remove them is to use `rowMeans` and `rowMedians`

```
low_genes = rowMeans(edata) < 5
table(low_genes)
```

```
## low_genes
## FALSE  TRUE
## 11413 41167
```

```
filt_edata = filter(edata,!low_genes)
dim(filt_edata)
```

```
## [1] 11413    19
```

```
low_genes2 = rowMedians(as.matrix(edata)) < 5
table(low_genes2,low_genes)
```

```
##          low_genes
## low_genes2 FALSE  TRUE
##      FALSE  9844    2
##      TRUE   1569 41165
```



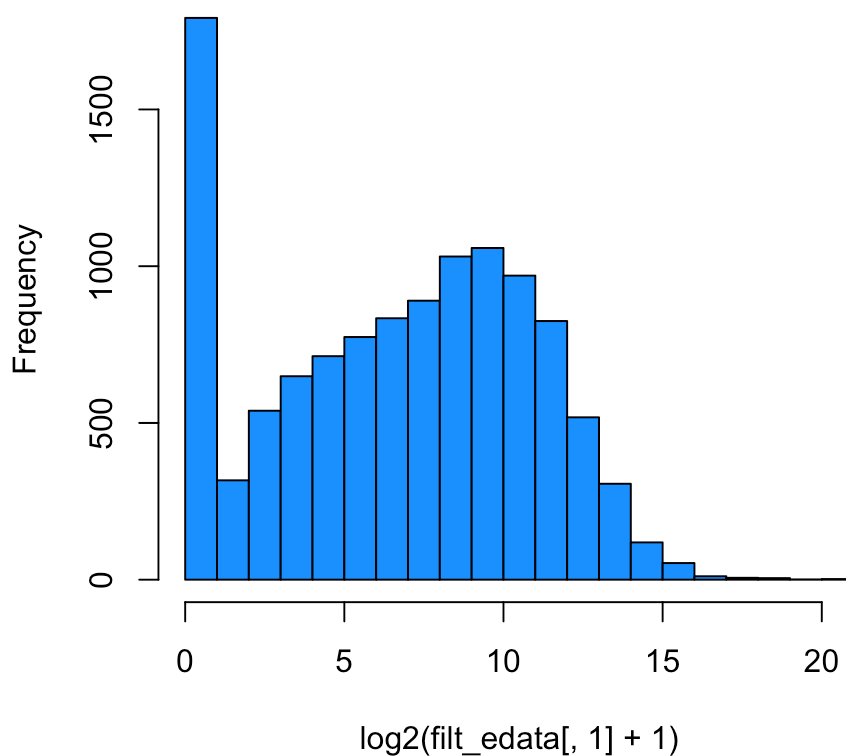
```
filt_edata2 = filter(edata,!low_genes2)
dim(filt_edata2)
```

```
## [1] 9846 19
```

After filtering we observe that using the `log` transform makes the distribution much more symmetric but there are still zeros. We could increase the filter to remove this issue if we needed to.

```
hist(log2(filt_edata[,1] + 1),col=2)
```

Histogram of $\log_2(\text{filt_edata}[, 1] + 1)$



More transforms

There are a large number of other transforms depending on the type of data you are using. Some common ones are:

- Variance stabilizing transforms (https://en.wikipedia.org/wiki/Variance-stabilizing_transformation) which seek to remove a mean variance relationship among the data
- Box-Cox transforms (https://en.wikipedia.org/wiki/Power_transform) which seek to make the data approximately Normally distributed
- rlog transform (<http://rpackages.ianhowson.com/bioc/DESeq2/man/rlog.html>) - unique to genomics count data, this is a regularized version of the log transform that seeks to minimize differences at low count levels.

Session information

Here is the session information

```
devtools::session_info()
```

```

## setting value
## version R version 3.2.1 (2015-06-18)
## system x86_64, darwin10.8.0
## ui RStudio (0.99.447)
## language (EN)
## collate en_US.UTF-8
## tz America/New_York
##
## package * version date
## acepack 1.3-3.3 2014-11-24
## annotate 1.46.1 2015-07-11
## AnnotationDbi * 1.30.1 2015-04-26
## assertthat 0.1 2013-12-06
## BiasedUrn * 1.06.1 2013-12-29
## Biobase * 2.28.0 2015-04-17
## BiocGenerics * 0.14.0 2015-04-17
## BiocInstaller * 1.18.4 2015-07-22
## BiocParallel 1.2.20 2015-08-07
## biomaRt 2.24.0 2015-04-17
## Biostrings 2.36.3 2015-08-12
## bitops 1.0-6 2013-08-17
## bladderbatch * 1.6.0 2015-08-26
## broom * 0.3.7 2015-05-06
## caTools 1.17.1 2014-09-10
## cluster 2.0.3 2015-07-21
## colorspace 1.2-6 2015-03-11
## corpcor 1.6.8 2015-07-08
## curl 0.9.2 2015-08-08
## DBI * 0.3.1 2014-09-24
## dendextend * 1.1.0 2015-07-31
## DESeq2 * 1.8.1 2015-05-02
## devtools * 1.8.0 2015-05-09
## digest 0.6.8 2014-12-31
## dplyr * 0.4.3 2015-09-01
## edge * 2.1.0 2015-09-06
## evaluate 0.7.2 2015-08-13
## foreign 0.8-66 2015-08-19
## formatR 1.2 2015-04-21
## Formula * 1.2-1 2015-04-07
## futile.logger 1.4.1 2015-04-20
## futile.options 1.0.0 2010-04-06
## gdata 2.17.0 2015-07-04
## genefilter * 1.50.0 2015-04-17
## geneLenDataBase * 1.4.0 2015-09-06
## geneplotter 1.46.0 2015-04-17
## GenomeInfoDb * 1.4.2 2015-08-15
## GenomicAlignments 1.4.1 2015-04-24
## GenomicFeatures 1.20.2 2015-08-14
## GenomicRanges * 1.20.5 2015-06-09
## genstats * 0.1.02 2015-09-05
## ggplot2 * 1.0.1 2015-03-17
## git2r 0.11.0 2015-08-12

```

##	GO.db	3.1.2	2015-09-06
##	goseq	* 1.20.0	2015-04-17
##	gplots	* 2.17.0	2015-05-02
##	gridExtra	2.0.0	2015-07-14
##	gtable	0.1.2	2012-12-05
##	gtools	3.5.0	2015-05-29
##	highr	0.5	2015-04-21
##	HistData	* 0.7-5	2014-04-26
##	Hmisc	* 3.16-0	2015-04-30
##	htmltools	0.2.6	2014-09-08
##	httr	1.0.0	2015-06-25
##	IRanges	* 2.2.7	2015-08-09
##	KernSmooth	2.23-15	2015-06-29
##	knitr	* 1.11	2015-08-14
##	lambda.r	1.1.7	2015-03-20
##	lattice	* 0.20-33	2015-07-14
##	latticeExtra	0.6-26	2013-08-15
##	lazyeval	0.1.10	2015-01-02
##	limma	* 3.24.15	2015-08-06
##	lme4	1.1-9	2015-08-20
##	locfit	1.5-9.1	2013-04-20
##	magrittr	1.5	2014-11-22
##	MASS	* 7.3-43	2015-07-16
##	Matrix	* 1.2-2	2015-07-08
##	MatrixEQTL	* 2.1.1	2015-02-03
##	memoise	0.2.1	2014-04-22
##	mgcv	* 1.8-7	2015-07-23
##	minqa	1.2.4	2014-10-09
##	mnormt	1.5-3	2015-05-25
##	munsell	0.4.2	2013-07-11
##	nlme	* 3.1-122	2015-08-19
##	nloptr	1.0.4	2014-08-04
##	nnet	7.3-10	2015-06-29
##	org.Hs.eg.db	* 3.1.2	2015-07-17
##	plyr	1.8.3	2015-06-12
##	preprocessCore	* 1.30.0	2015-04-17
##	proto	0.3-10	2012-12-22
##	psych	1.5.6	2015-07-08
##	qvalue	* 2.0.0	2015-04-17
##	R6	2.1.1	2015-08-19
##	RColorBrewer	1.1-2	2014-12-07
##	Rcpp	* 0.12.0	2015-07-25
##	RcppArmadillo	* 0.5.400.2.0	2015-08-17
##	RCurl	1.95-4.7	2015-06-30
##	reshape2	1.4.1	2014-12-06
##	rmarkdown	0.7	2015-06-13
##	rpart	4.1-10	2015-06-29
##	Rsamtools	1.20.4	2015-06-01
##	RSkittleBrewer	* 1.1	2015-09-05
##	RSQLite	* 1.0.0	2014-10-25
##	rstudioapi	0.3.1	2015-04-07
##	rtracklayer	1.28.9	2015-08-19

##	rversions	1.0.2	2015-07-13
##	S4Vectors	* 0.6.5	2015-09-01
##	scales	0.3.0	2015-08-25
##	snm	1.16.0	2015-04-17
##	snpStats	* 1.18.0	2015-04-17
##	stringi	0.5-5	2015-06-29
##	stringr	1.0.0	2015-04-30
##	survival	* 2.38-3	2015-07-02
##	sva	* 3.14.0	2015-04-17
##	tidyr	0.2.0	2014-12-05
##	UsingR	* 2.0-5	2015-08-06
##	whisker	0.3-2	2013-04-28
##	XML	3.98-1.3	2015-06-30
##	xml2	0.1.2	2015-09-01
##	xtable	1.7-4	2014-09-12
##	XVector	0.8.0	2015-04-17
##	yaml	2.1.13	2014-06-12
##	zlibbioc	1.14.0	2015-04-17
##	source		
##	CRAN (R 3.2.0)		
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##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.2)		
##	Github (jdstorey/edge@a1947b5)		
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## CRAN (R 3.2.1)
## Bioconductor
## Github (alyssafranze/RSkittleBrewer@0a96a20)
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## CRAN (R 3.2.2)
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## Bioconductor
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It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.